

Endterm Examination =

Solutions

1. Using Bessel's inequality we have,

$$\sum_{n=1}^{\infty} |\langle y, e_n \rangle|^2 \leq \|y\|^2 < \infty \quad (\text{for each } y)$$

$$\Rightarrow z_n := \langle y, e_n \rangle \Rightarrow \sum |z_n|^2 < \infty$$

$$\Rightarrow z_n \rightarrow 0 \quad (\text{nth term of the seq})$$

$$\Rightarrow \langle y, e_n \rangle \rightarrow 0 \quad \text{for each } y.$$

2. Given, $\langle x_n, x \rangle \rightarrow \|x\|^2 \Rightarrow \langle x, x_n \rangle = \overline{\langle x_n, x \rangle}$

$$, \quad \|x_n\| \rightarrow \|x\|$$

$$\downarrow$$
$$\underbrace{\overline{\|x\|^2}}_{\in \mathbb{R}} = \|x\|^2.$$

$$\Rightarrow \langle x, x_n \rangle \rightarrow \|x\|^2 \text{ also.}$$

$$\|x_n - x\|^2 = \langle x_n - x, x_n - x \rangle$$

$$= \|x_n\|^2 - \langle x, x_n \rangle - \langle x_n, x \rangle + \|x\|^2$$

$$\rightarrow \|x\|^2 - \|x\|^2 - \|x\|^2 + \|x\|^2 = 0.$$

$$\Rightarrow \|x_n - x\|^2 \rightarrow 0 \text{ in } \mathbb{R}.$$

4. Let $\{e_n\}$ be an orthonormal basis in a Hilbert space.

Let us assume that the set has k^2+1 elements.

Since, $\{e_n\}_n$ - orthonormal Basis, we have for any

$$x \in X, \quad \sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2 = \|x\|^2$$

$$1 = \|x\|^2 = \sum_{i=1}^{\infty} |\langle x, e_i \rangle|^2 \geq \frac{1}{k^2} (k^2+1) > 1.$$

A contradiction.

Hence, the set $\{i \in \mathbb{N} : |\langle x, e_i \rangle| \geq \frac{1}{k}\}$ has at most k^2 -element.

5. Consider the seq $\{x_n = \frac{1}{n}\}$. Clearly $\{x_n\}$ is a Cauchy

$$\text{seq. } d(x_n, x_m) = \left| \frac{1}{x_n} - \frac{1}{x_m} \right| = \left| \frac{1}{n} - \frac{1}{m} \right| < \varepsilon \quad \forall \quad \underline{n, m \geq N_0}.$$

If it is complete, then it must converge

to some m_0 .

$$\Rightarrow \{x_n = \frac{1}{n}\} \text{ has no limit in } \mathbb{N} = m_0^{-1} > 0. \rightarrow \text{f.}$$

6. $\tau = \{U \in \mathcal{R} : \mathbb{R} \setminus U = \emptyset \text{ or } \mathbb{R} \setminus U \text{ is infinite}\}$

Take $U = (0, \infty)$, $V = (-\infty, 0)$

$\mathbb{R} \setminus U = (-\infty, 0]$ & $\mathbb{R} \setminus V = [0, \infty)$, both of which are infinite.

Hence $U, V \in \tau$. But $U \cup V = \mathbb{R} \setminus \{0\}$

$\Rightarrow \mathbb{R} \setminus (U \cup V) = \{0\}$

Therefore, $U \cup V \notin \tau$.

$\Rightarrow \tau$ is not a topology on $\mathbb{R}$.

↑
which is finite
singleton set &
non-empty.

7. $X = \mathbb{R}$, $A = [0, 1]$, $B = [2, 3]$. Both are

with
standard
topo.

$A \cup B = [0, 1] \cup [2, 3]$



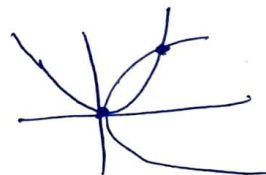
is disconnected

$X = \mathbb{R}^2$

$A = \{(x, y) : y = x^2\}$, $B = \{(x, y) : x = y^2\}$

A, B are connected sets in $\mathbb{R}^2$.

$A \cap B = \{0, 1\}$ - disconnected.



9. $f: (0,1) \rightarrow \mathbb{R}$, $f(x) = \sqrt{x} \log x$.

$$\lim_{x \rightarrow 1^-} f(x) = 0 , \quad \lim_{x \rightarrow 0^+} f(x) = 0 .$$

Define $\tilde{f}(x) = \begin{cases} f(x) & , 0 < x < 1 \\ 0 & , x = 0, 1 . \end{cases}$

Clearly $\tilde{f}$ is ~~unif~~ continuous function in $[0,1]$.

Since $[0,1]$ is closed & bounded in $\mathbb{R}$ & hence cpt

$\Rightarrow \tilde{f}$ in $[0,1]$ is uniformly continuous .

$\Rightarrow f = \tilde{f}|_{(0,1)}$ is also uniformly continuous .

8. Assume $d(A,B) = 0$. By the definition $\exists \{a_n\} \in A$ & $\{b_n\} \in B$ s.t.

$$d(a_n, b_n) < \frac{1}{n} \Rightarrow d(a_n, b_n) \rightarrow 0 \text{ as } n \rightarrow \infty .$$

Since B is compact, $\{b_n\}$ has a convergent

subseq $\{b_{n_k}\}$ i.e., $b_{n_k} \rightarrow b \in B$, $k \rightarrow \infty$

$$0 \leq d(a_{n_k}, b) \leq d(a_{n_k}, b_{n_k}) + d(b_{n_k}, b) \rightarrow 0 + 0 = 0 .$$

$$\Rightarrow a_{n_k} \rightarrow b \text{ as } k \rightarrow \infty$$

$\Rightarrow b$ is a limit point of $\{a_{n_k}\} \subseteq A$. Since

A is closed $\Rightarrow b \in A$. ~~but~~ $\Rightarrow b \in \underbrace{A \cap B}_{\neq \emptyset} \rightarrow$

$$\Rightarrow \underline{d(A,B) > 0} ,$$

3. Let $x \in (M+N)^\perp$

$$M+N = \{y+z : y \in M, z \in N\}$$

$$\Rightarrow \langle x, y+z \rangle = 0 \quad \forall y \in M, z \in N$$

$$\Rightarrow \langle x, y \rangle + \langle x, z \rangle = 0 \quad \forall y \in M, z \in N$$

Since M, N are subspaces $\Rightarrow 0 \in M$ & N

In particular, ^{Choosing} $z = 0 \Rightarrow \langle x, y \rangle = 0$

^{||y} $y = 0 \Rightarrow \langle x, z \rangle = 0$

Hence, $x \in M^\perp$ & $x \in N^\perp \Rightarrow x \in M^\perp \cap N^\perp$

$$\Rightarrow (M+N)^\perp \subseteq M^\perp \cap N^\perp \quad \rightarrow (1)$$

Conversely, $x \in M^\perp \cap N^\perp$, let $y+z \in M+N$, $y \in M, z \in N$

Consider,
$$\begin{aligned} \langle x, y+z \rangle &= \underbrace{\langle x, y \rangle}_0 + \underbrace{\langle x, z \rangle}_0 \\ &= 0 \end{aligned}$$

$$\Rightarrow \langle x, y+z \rangle = 0 \Rightarrow x \in (M+N)^\perp$$

$$\Rightarrow M^\perp \cap N^\perp \subseteq (M+N)^\perp \quad \rightarrow (2)$$

(1) & (2)

$$\Rightarrow M^\perp \cap N^\perp = (M+N)^\perp$$